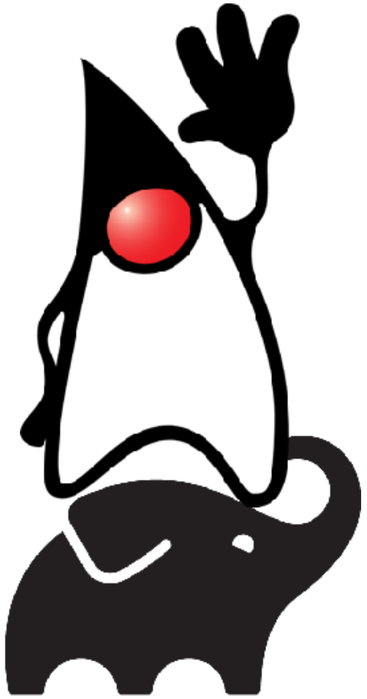
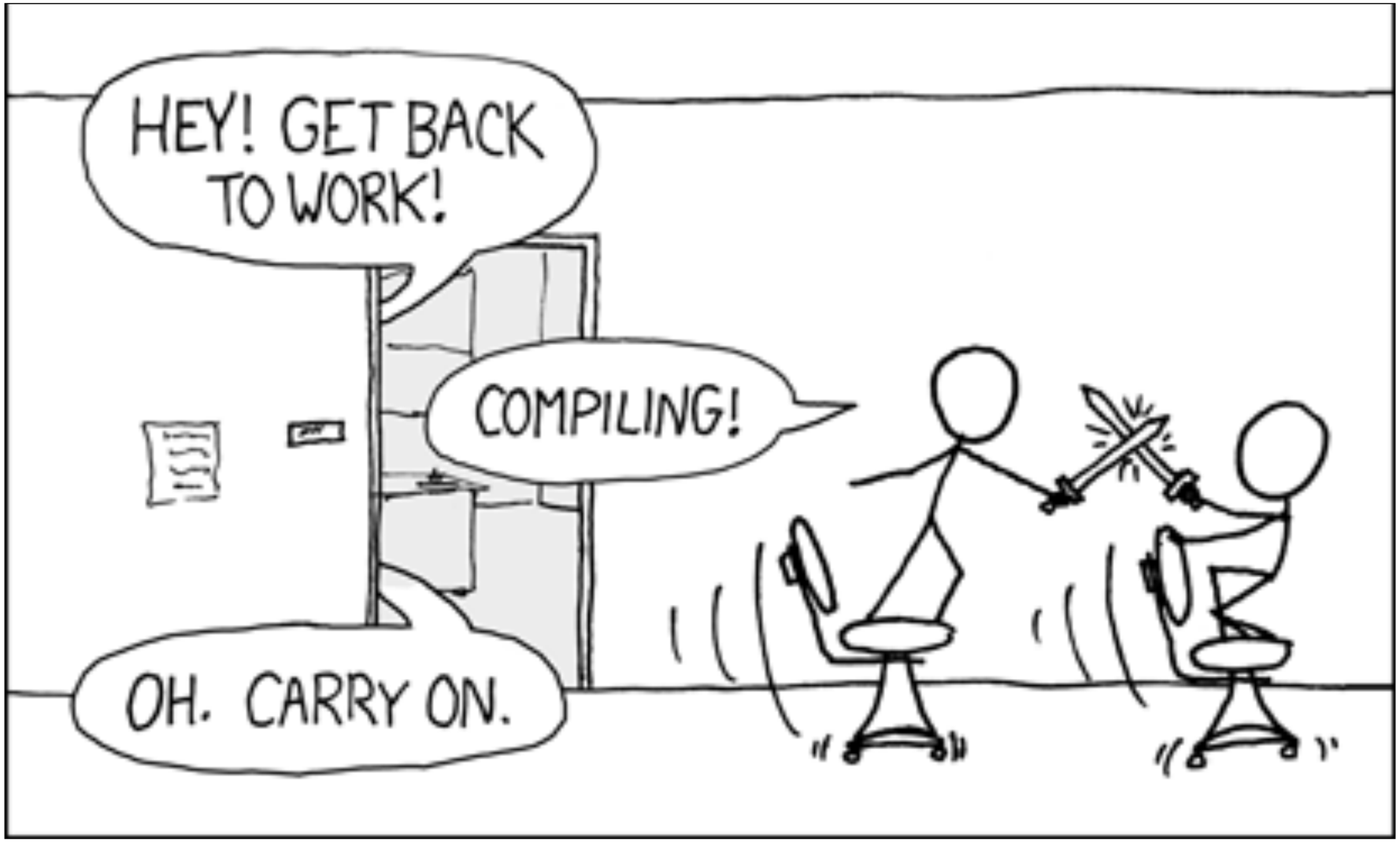
Combining Java Modules and Gradle for elegant project structures



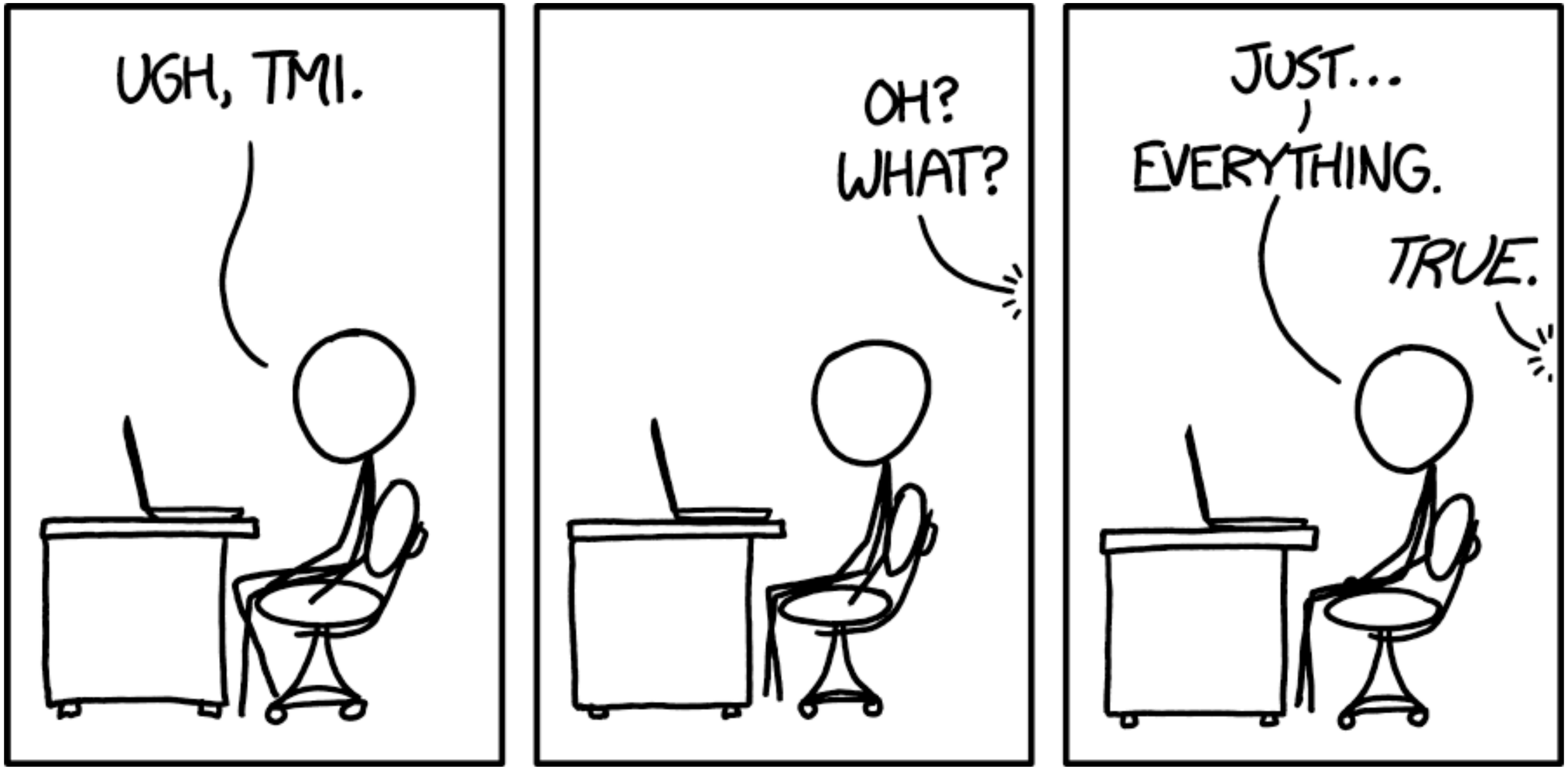
Jendrik Johannes

github.com/jjohannes

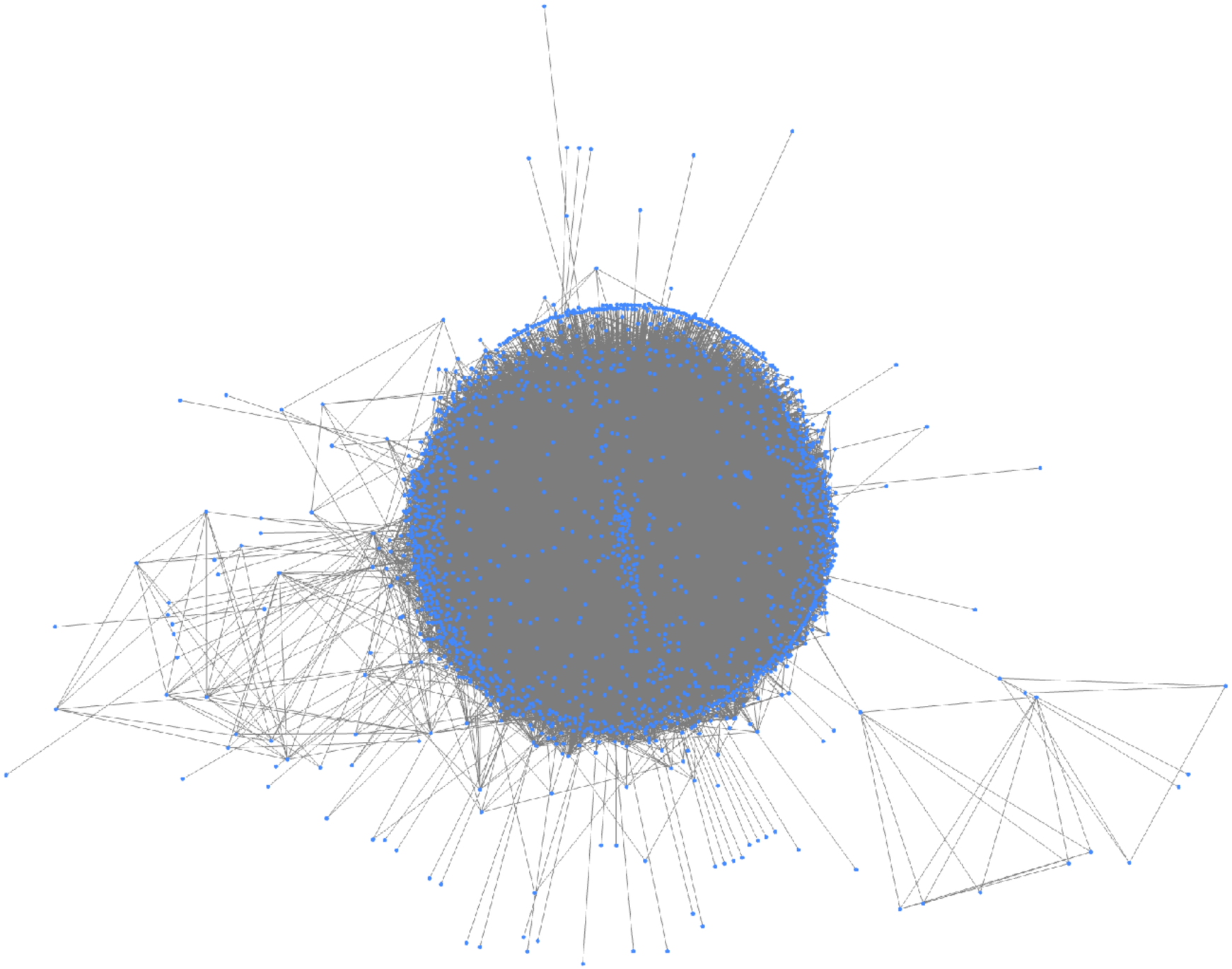
February 2025

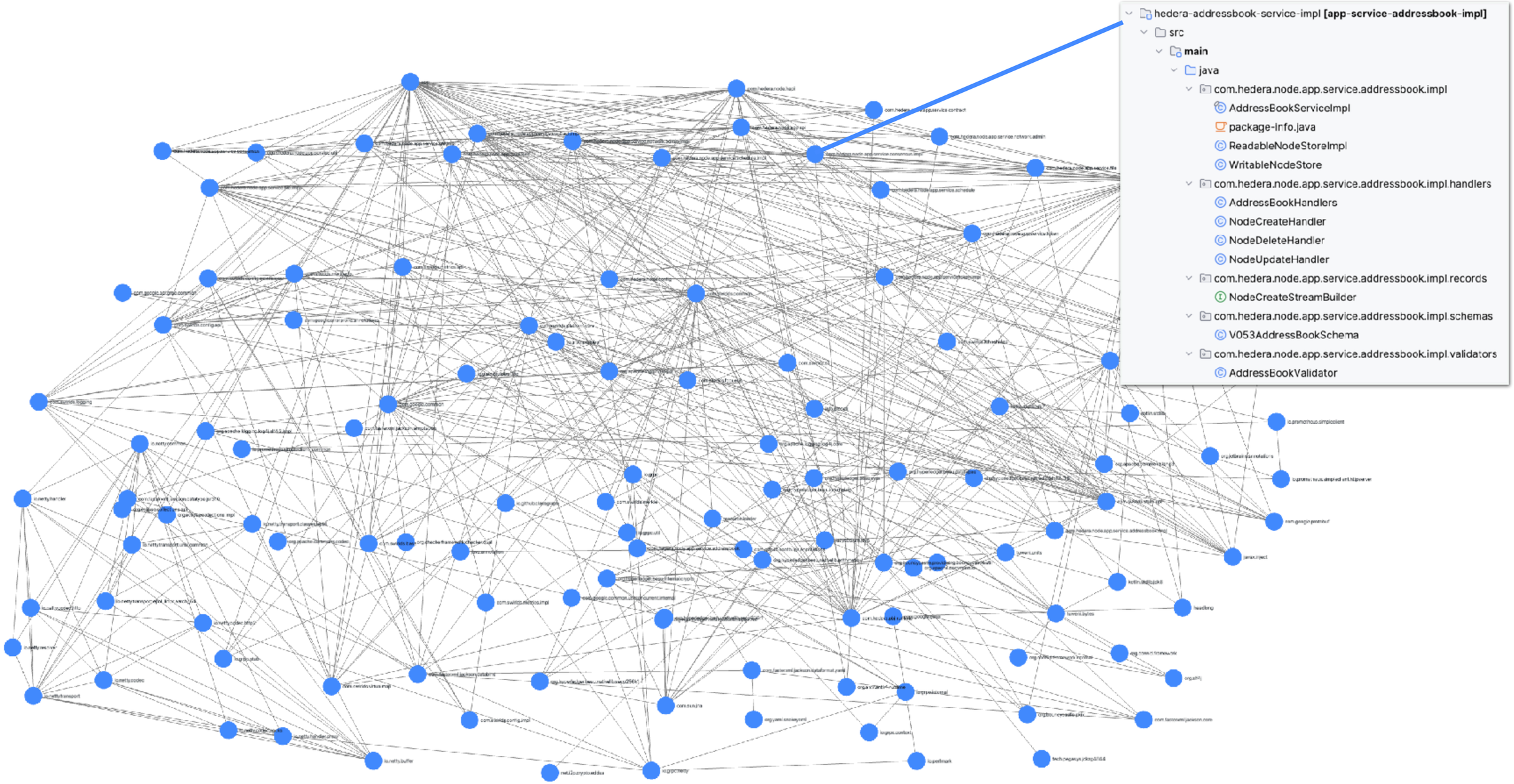


xkcd.com/303/



xkcd.com/1369/



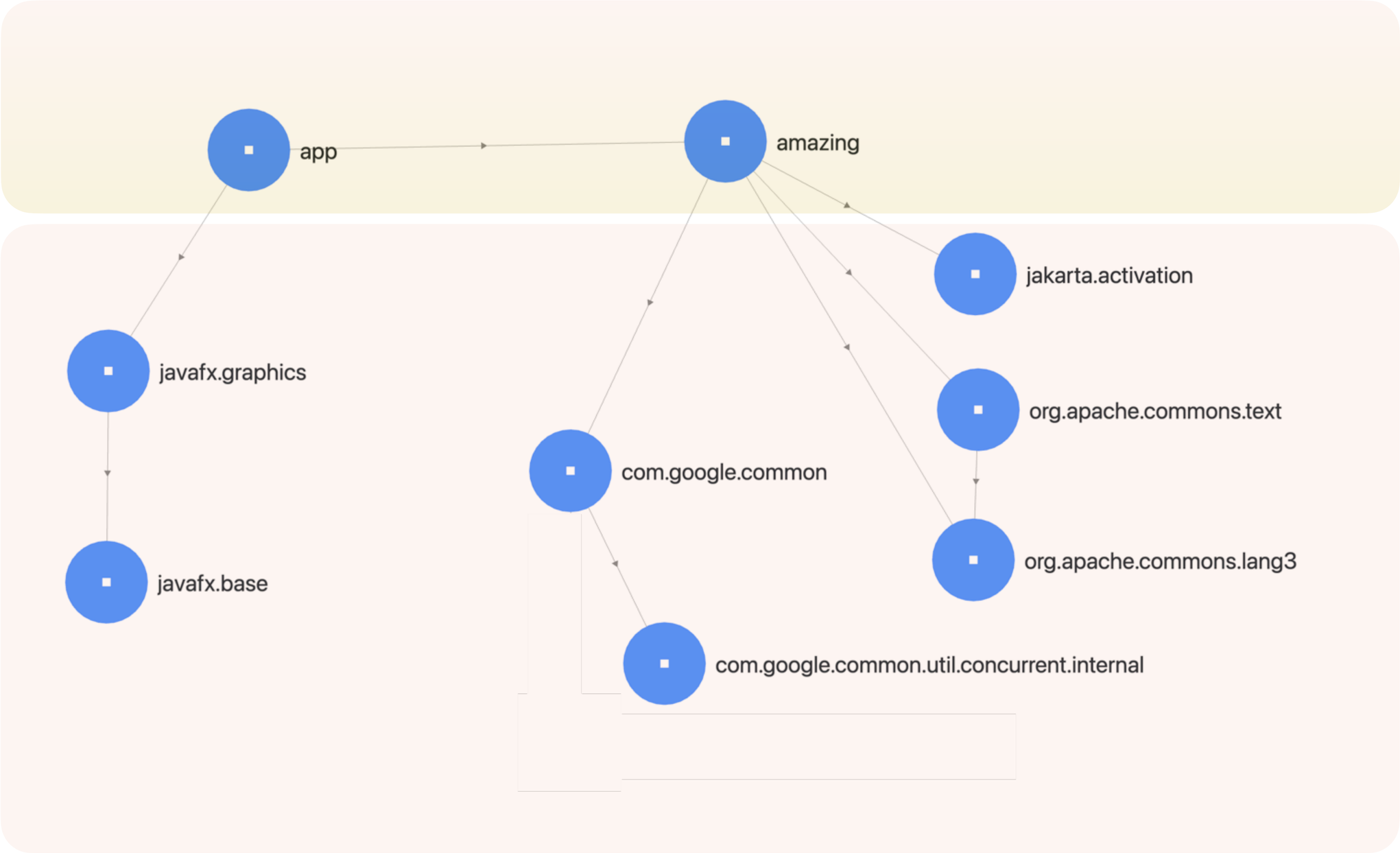


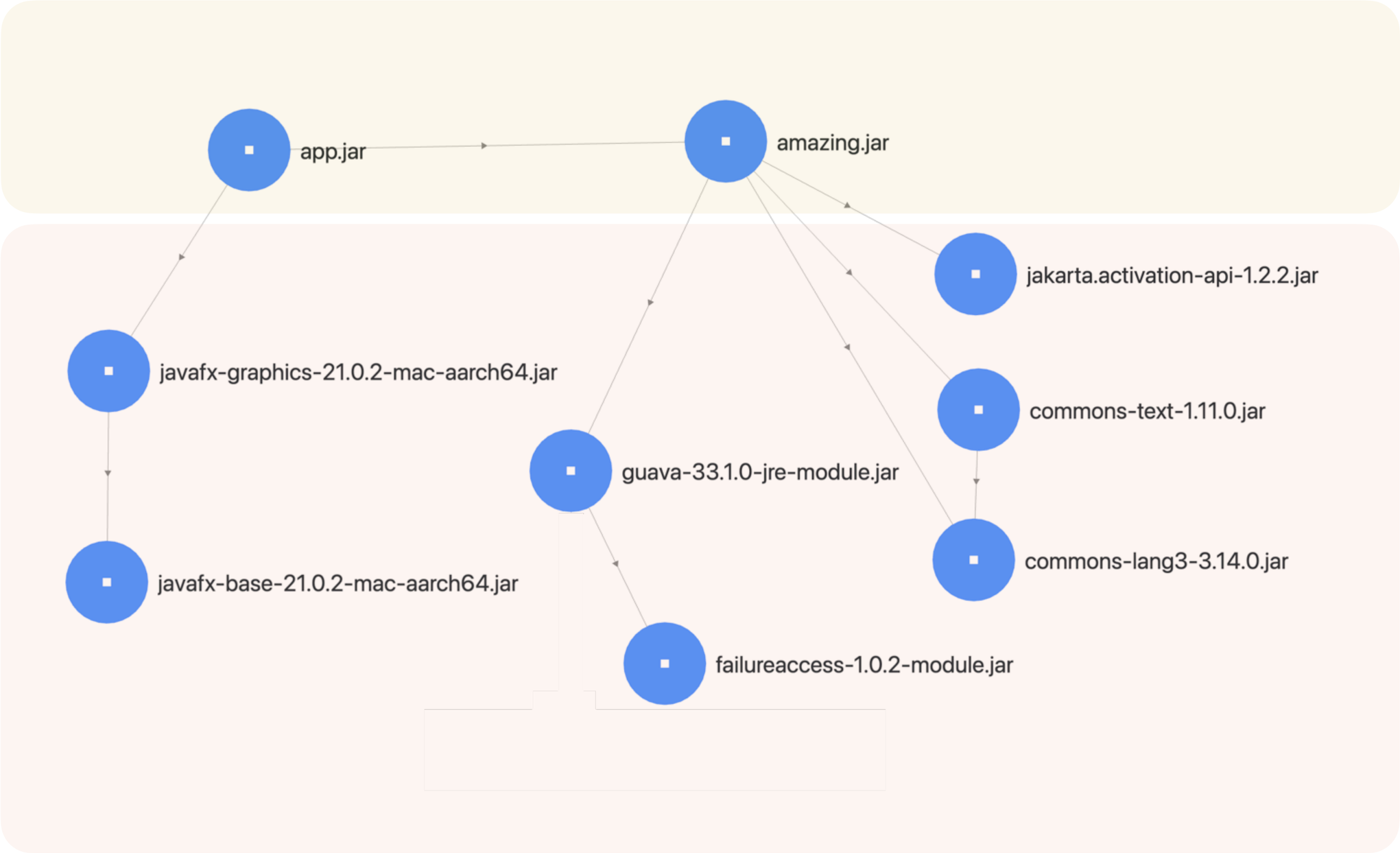
github.com/hiero-ledger/hiero-consensus-node

poloclub.github.io/argo-graph-lite/#d5b95db8-a363-46a5-8757-a846e116fa2b

Part 1

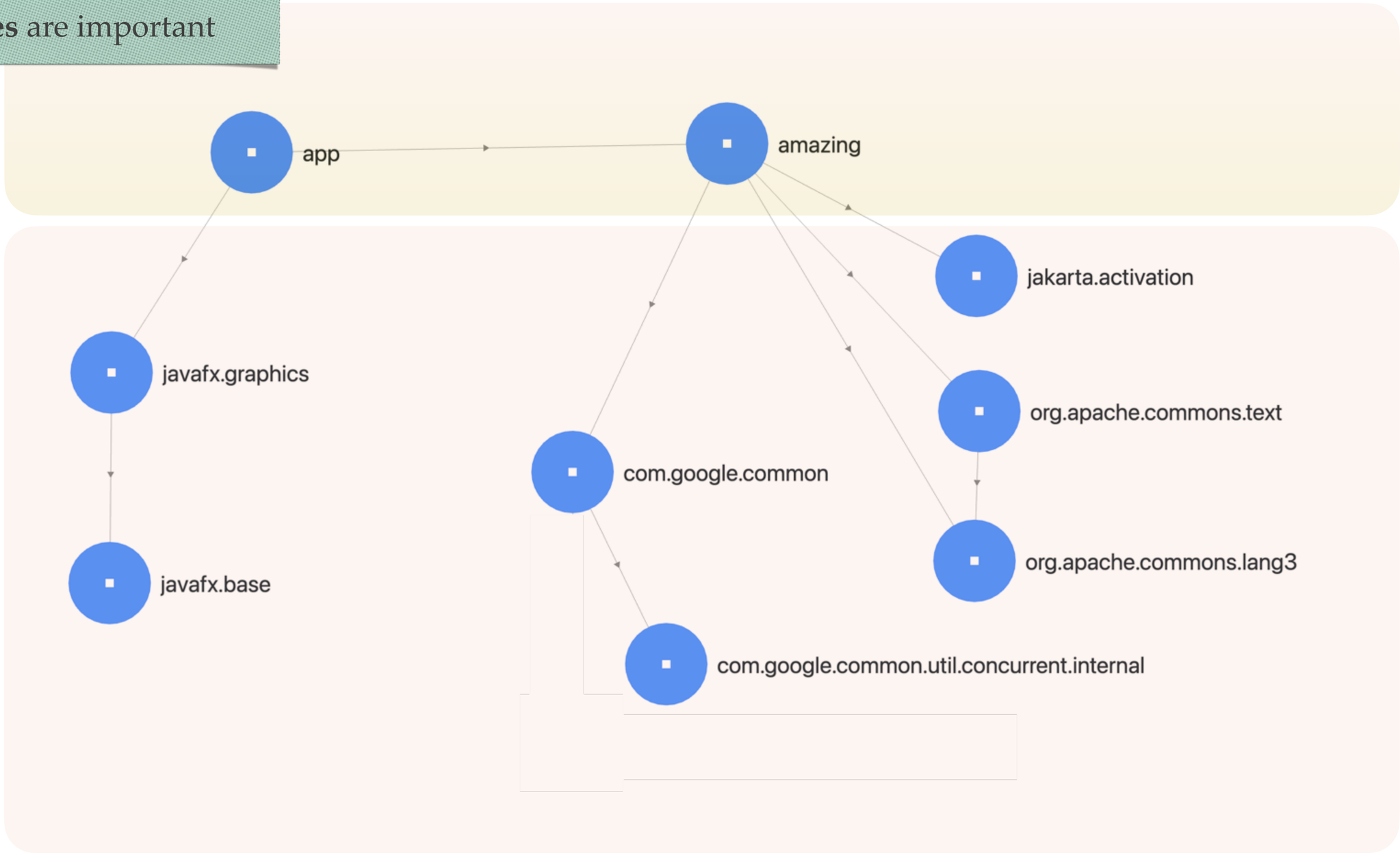
Complexity in **Module Definition**


app
amazing
jakarta.activation
javafx.graphics
org.apache.commons.text
com.google.common
org.apache.commons.lang3
javafx.base
com.google.common.util.concurrent.internal


app.jar
amazing.jar
jakarta.activation-api-1.2.2.jar
javafx-graphics-21.0.2-mac-aarch64.jar
commons-text-1.11.0.jar
guava-33.1.0-jre-module.jar
commons-lang3-3.14.0.jar
javafx-base-21.0.2-mac-aarch64.jar
failureaccess-1.0.2-module.jar

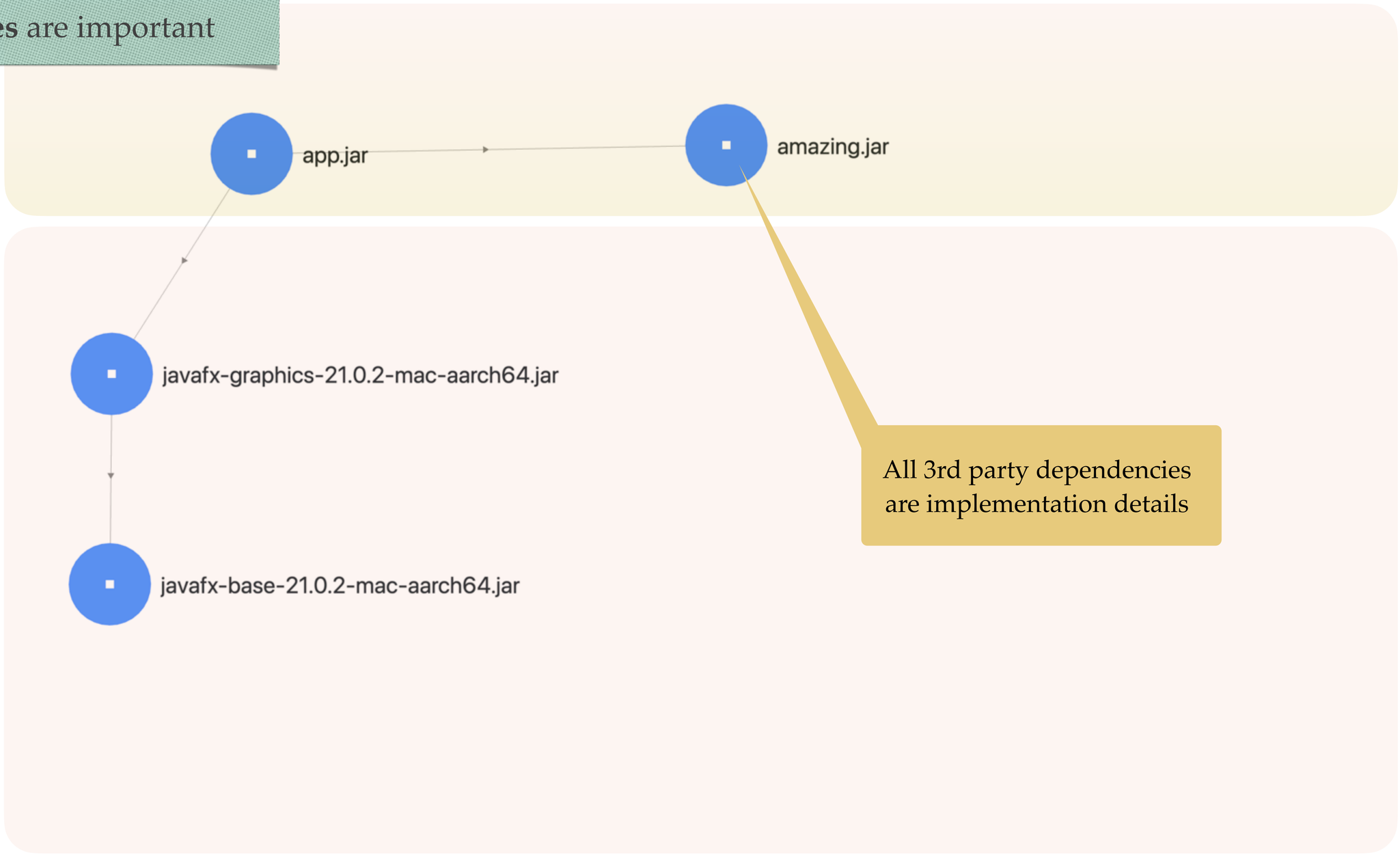
Essential Complexity #1

Scopes are important



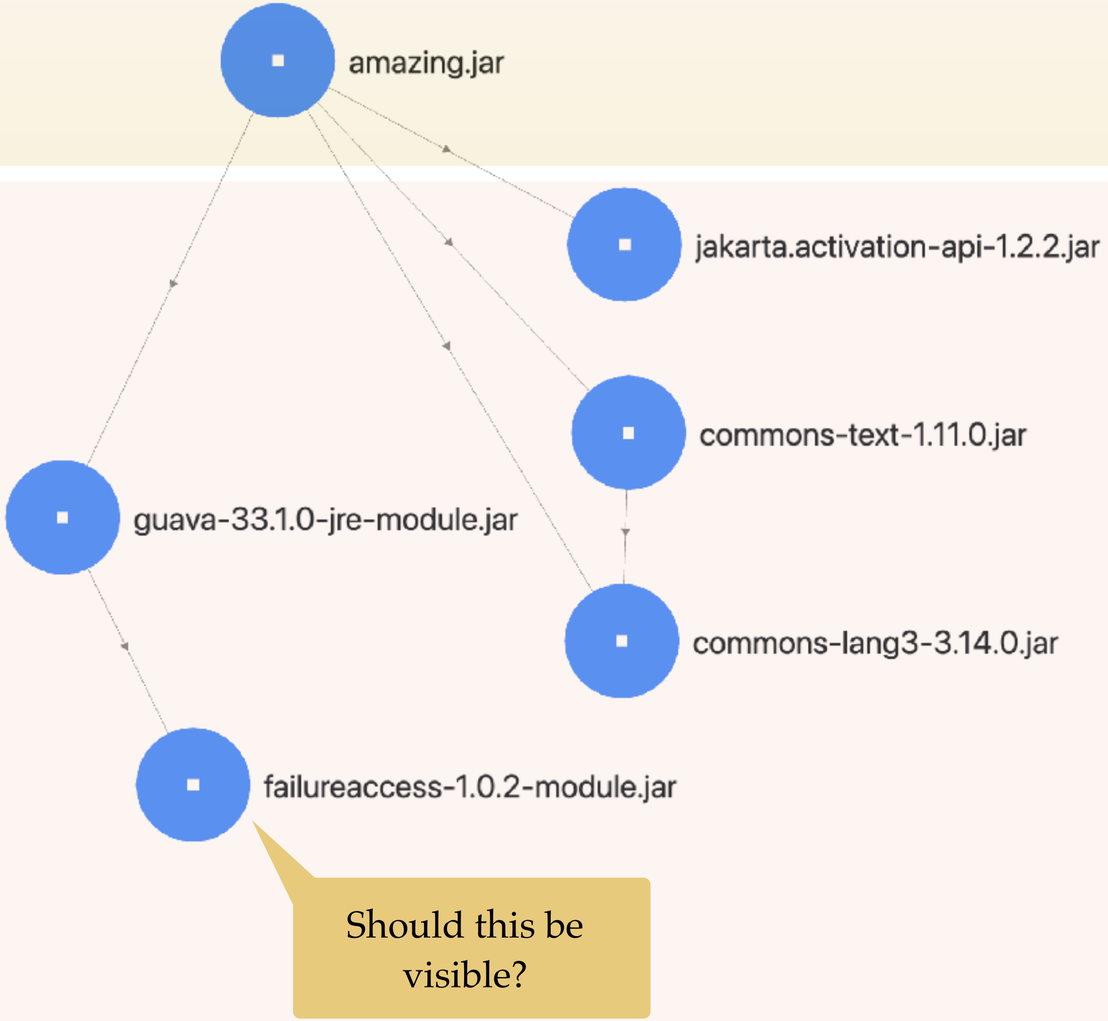
Essential Complexity #1

Scopes are important



Essential Complexity #1

Scopes are important

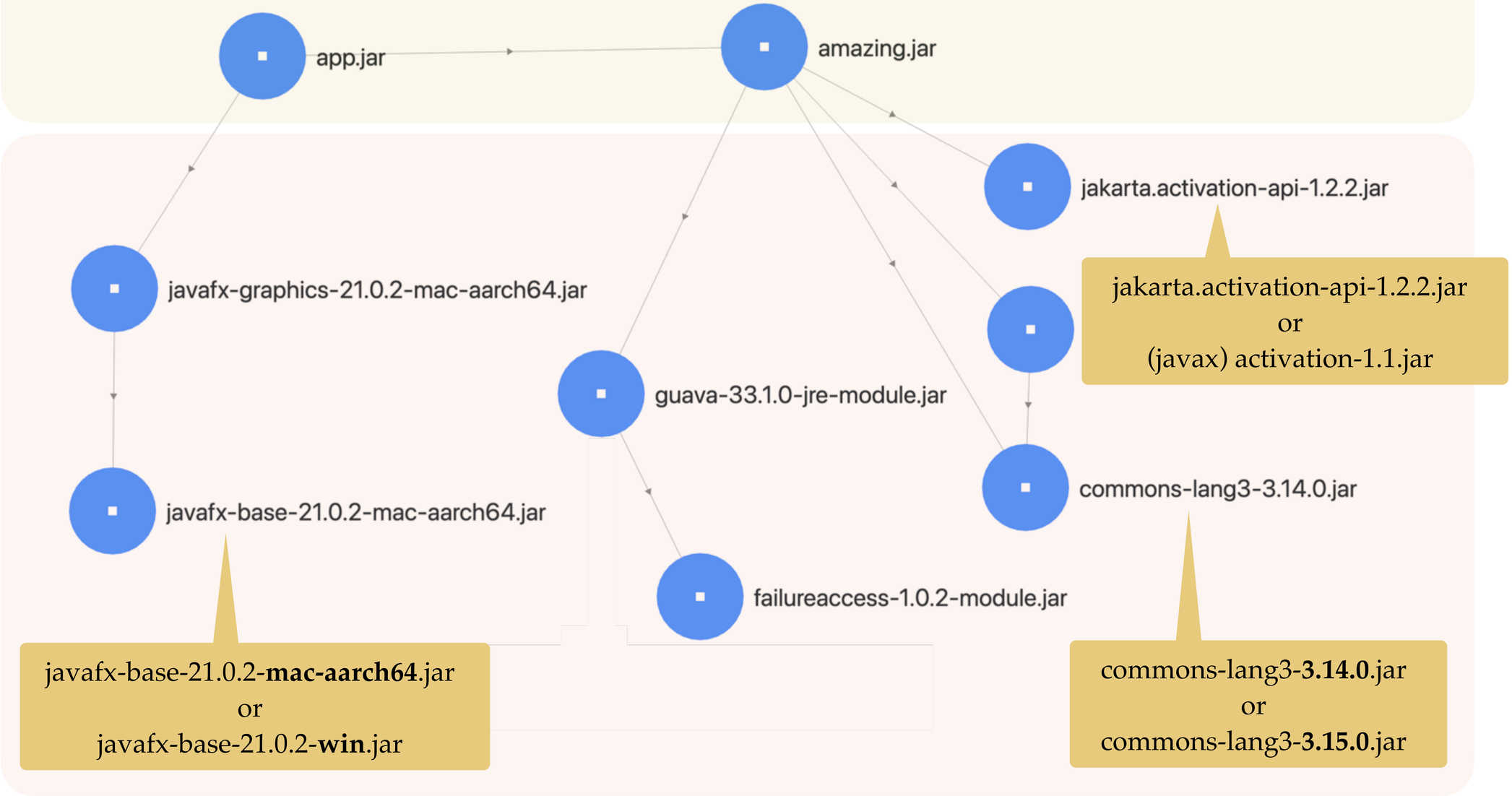


Essential Complexity #1

Scopes are important

Essential Complexity #2

Multiple **Versions/Variants** of 3rd party Modules



Essential Complexity #1

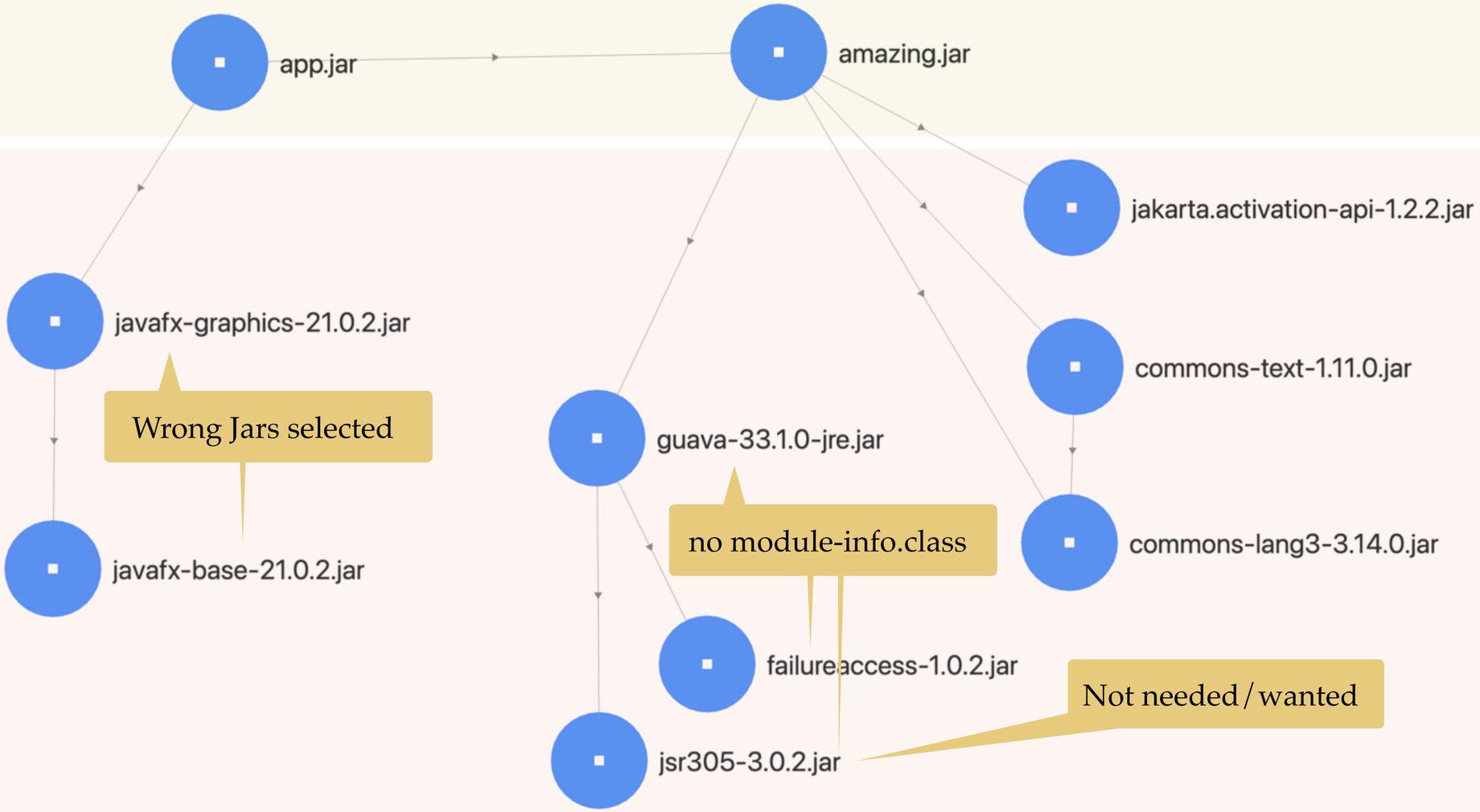
Scopes are important

Essential Complexity #2

Multiple **Versions/Variants** of 3rd party Modules

"Essential" Complexity #3

The world is **not perfect**



Part 2

Accidental Complexity
introduced through
insufficient Tooling and Knowledge

github.com/hibernate/hibernate-orm/blob/main/hibernate-core/hibernate-core.gradle

~300 Lines total
~20 Lines Module Definition

```
30  dependencies {
31      api jakartaLibs.jpa
32      api jakartaLibs.jta
33
34      implementation libs.hcann
35      implementation libs.jandex
36      implementation libs.classmate
37      implementation libs.byteBuddy
38
39      implementation jakartaLibs.jaxbApi
40      implementation jakartaLibs.jaxb
41      implementation jakartaLibs.inject
42
43      implementation libs.antlrRuntime
```

```
30  <dependencies>
31
32    <dependency>
33      <groupId>org.slf4j</groupId>
34      <artifactId>slf4j-api</artifactId>
35    </dependency>
36
37    <dependency>
38      <groupId>org.apache.hadoop</groupId>
39      <artifactId>hadoop-auth</artifactId>
40    </dependency>
41
42    <dependency>
43      <groupId>org.apache.hadoop</groupId>
44      <artifactId>hadoop-annotations</artifactId>
45    </dependency>
```

~300 Lines total
~100 Line Module Definition

github.com/apache/hadoop/blob/trunk/hadoop-common-project/hadoop-registry/pom.xml

Accidental Complexity #0
Module definition mixed with other build concerns

Maven Repository: org.apach

mvnrepository.com/artifact/org.apache.commons/commons-text/1.12.0

Home » org.apache.commons » commons-text » 1.12.0

Apache Commons Text » 1.12.0

The Commons Text library provides additions to the standard JDK text handling. It includes algorithms for string similarity and for calculating the distance between strings.

License	Apache 2.0
Categories	String Utilities
Tags	text string apache commons
HomePage	https://commons.apache.org/proper/commons-text
Date	Apr 16, 2024
Files	pom (19 KB) jar (245 KB) View All
Repositories	Central
Ranking	**#144 in MvnRepository (See Top Artifacts)** **#1 in String Utilities**
Used By	**3,599 artifacts**

Maven | Gradle | Gradle (Short) | Gradle (Kotlin) | SBT | Ivy | Grape | Leiningen | Buildr

```
<dependency>
    <groupId>org.apache.commons</groupId>
    <artifactId>commons-text</artifactId>
    <version>1.12.0</version>
</dependency>
```

Scope?

Accidental Complexity #1

Module definition mixed with other build concerns

Accidental Complexity #2

Scopes not cared about (enough)

Maven Repository: org.apach

mvnrepository.com/artifact/org.apache.commons/commons-text/1.12.0

Home » org.apache.commons » commons-text » 1.12.0

Apache Commons Text » 1.12.0

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```
implementation("org.apache.commons:commons-text:1.12.0")
```

api? runtimeOnly?

Accidental Complexity #0

Module definition mixed with other build concerns

Accidental Complexity #1

Scopes not cared about (enough)

repo1.maven.org/maven2/org/openjfx/javafx-graphics/21.0.2/

```
org/openjfx/javafx-graphics/21.0.2/
├── javafx-graphics-21.0.2.pom
├── javafx-graphics-21.0.2.module
├── javafx-graphics-21.0.2.jar
├── javafx-graphics-21.0.2-linux.jar
│   └── module-info.class
├── javafx-graphics-21.0.2-mac.jar
│   └── module-info.class
└── javafx-graphics-21.0.2-win.jar
    └── module-info.class
```

No rich metadata beyond POM 4.0.0 👎

Empty Jar to get Maven to do a OS-based selection
-> such hacks shouldn't be needed
-> does not work for Gradle 👎

Java Module Descriptor
(Java Module System / JPMS)
"Java standard notation" 👍

Accidental Complexity #0
Module definition mixed with other build concerns

Accidental Complexity #1
Scopes not cared about (enough)

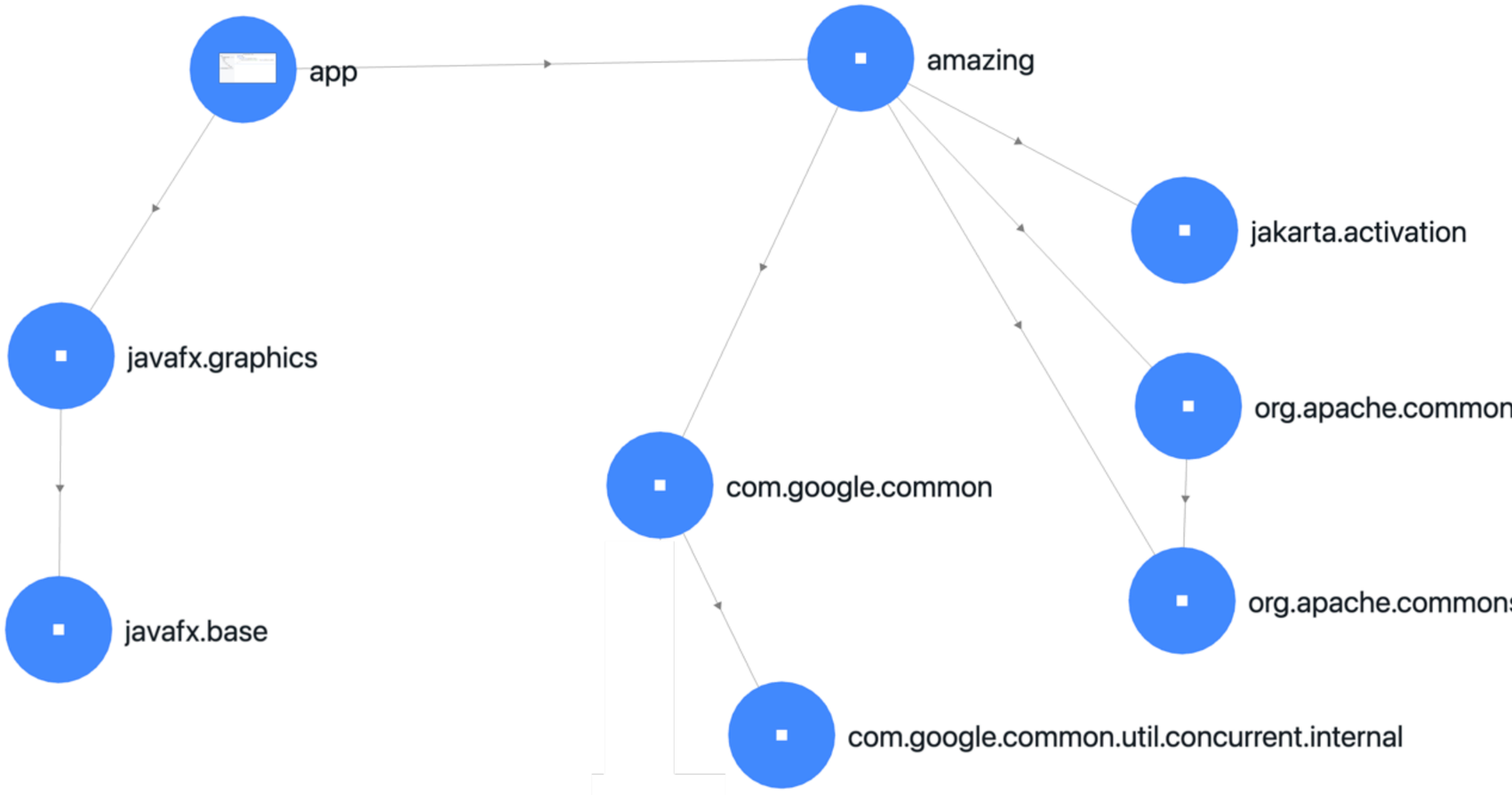
Accidental Complexity #2
Multiple Version/Variants not cared about (enough)

Accidental Complexity #3
Metadata of 3rd party modules is incomplete or wrong

Part 3

How to do

Better?


app
amazing
jakarta.activation
javafx.graphics
org.apache.common
com.google.common
org.apache.commons
javafx.base
com.google.common.util.concurrent.internal

java-module-system – build.gradle.kts (:amazing)

Project

- java-module-system ~/projects/gradle/hov
  - gradle
  - modules
    - amazing [java-module-system.amazi
      - src
        - main
          - java
            - amazing
      - build.gradle.kts
    - app [java-module-system.app]
      - src
        - main
          - java
            - app
      - build.gradle.kts
  - settings.gradle.kts

build.gradle.kts (:amazing)

```
dependencies {
    implementation("com.google.guava:guava")
    implementation("org.apache.commons:commons-text")
}
```

build.gradle.kts (:app)

```
dependencies {
    implementation(project(":amazing"))
    implementation("org.openjfx:javafx-graphics")
}
```

Module Definition

java-module-system – compile-and-test.gradle.kts [plugins.main]

Project

- java-module-system ~/projects/gradle/howto/java-moc
  - gradle
    - plugins
      - src
        - main
          - kotlin
            - build-performance.settings.gradle.kts
            - compile-and-test.gradle.kts
            - dependency-analysis.gradle.kts
            - java-module.gradle.kts
            - metadata-patch.gradle.kts
            - module-locations.settings.gradle.kts
            - targets.gradle.kts
        - build.gradle.kts
    - versions [java-module-system.versions]
      - build.gradle.kts
  - modules
  - settings.gradle.kts

```kotlin
plugins {
    id("java")
    id("org.gradlex.java-module-testing")
}

java {
    toolchain.languageVersion = JavaLanguageVersion.of(21)
}
tasks.withType<JavaCompile>().configureEach {
    options.encoding = "UTF-8"
}
tasks.withType<Test>().configureEach {
    maxParallelForks = 4
    maxHeapSize = "1g"
}
```

Other Build Concerns

java-module-system – metadata-patch.gradle.kts [plugins.main]

Project

- java-module-system ~/projects/gradle/howto/
  - gradle
    - plugins
      - src
        - main
          - kotlin
            - build-performance.settings.gr
            - compile-and-test.gradle.kts
            - dependency-analysis.gradle.k
            - java-module.gradle.kts
            - metadata-patch.gradle.kts
            - module-locations.settings.gra
            - targets.gradle.kts
      - build.gradle.kts
    - versions [java-module-system.versions]
      - build.gradle.kts
  - modules
  - settings.gradle.kts

```kotlin
jvmDependencyConflicts { // Patch Maven/Gradle metadata (*.pom / *.module)
    patch {
        module("com.google.guava:guava") {
            removeDependency("com.google.code.findbugs:jsr305")
            removeDependency("org.checkerframework:checker-qual")
            removeDependency("com.google.errorprone:error_prone_annotations")
        }
        listOf("base", "graphics", "controls").forEach { jfxModule ->
            module("org.openjfx:javafx-$jfxModule") {
                addTargetPlatformVariant("linux", OperatingSystemFamily.LINUX
                addTargetPlatformVariant("mac", OperatingSystemFamily.MACOS,
                addTargetPlatformVariant("mac-aarch64", OperatingSystemFamily
                addTargetPlatformVariant("win", OperatingSystemFamily.WINDOWS
            }
        }
    }
}
```

Module Patching

java-module-system – build.gradle.kts (:amazing)

Project

- java-module-system ~/projects/gradle/hov
  - gradle
  - modules
    - amazing [java-module-system.amazi
      - src
        - main
          - java
            - amazing
      - build.gradle.kts
    - app [java-module-system.app]
      - src
        - main
          - java
            - app
      - build.gradle.kts
  - settings.gradle.kts

build.gradle.kts (:amazing)

```
dependencies {
    implementation("com.google.guava:guava")
    implementation("org.apache.commons:commons-text")
}
```

build.gradle.kts (:app)

```
dependencies {
    implementation(project(":amazing"))
    implementation("org.openjfx:javafx-graphics")
}
```

Module Definition

java-module-system – module-info.java (app)

Project

- java-module-system ~/projects/gradle/hov
  - gradle
  - modules
    - amazing [java-module-system.amazi
      - src
        - main
          - java
            - amazing
              - module-info.java
    - app [java-module-system.app]
      - src
        - main
          - java
            - app
              - module-info.java
  - settings.gradle.kts

module-info.java (amazing)

```
module amazing {
    requires com.google.common;
    requires org.apache.commons.text;

    exports amazing;
}
```

module-info.java (app)

```
module app {
    requires amazing;
    requires javafx.graphics;

    exports app;
}
```

Module Definition

java-module-system – build.gradle.kts (:plugins)

Project

- java-module-system ~/projects/grad
  - gradle
    - plugins
      - src
      - build.gradle.kts
    - versions [java-module-system
      - build.gradle.kts
  - modules
  - settings.gradle.kts

```kotlin
plugins {
    `kotlin-dsl`
}

repositories {
    gradlePluginPortal()
}

dependencies {
    implementation("com.autonomousapps:dependency-analysis-gradle-plugin:1.32.0")
    implementation("com.gradle:develocity-gradle-plugin:3.17.5")
    implementation("org.gradlex:extra-java-module-info:1.8")
    implementation("org.gradlex:java-module-dependencies:1.7")
    implementation("org.gradlex:java-module-packaging:0.1")
    implementation("org.gradlex:java-module-testing:1.4")
    implementation("org.gradlex:jvm-dependency-conflict-resolution:2.1.1")
}
```

Plugins for Module Management

Summary

Essential Complexity #1

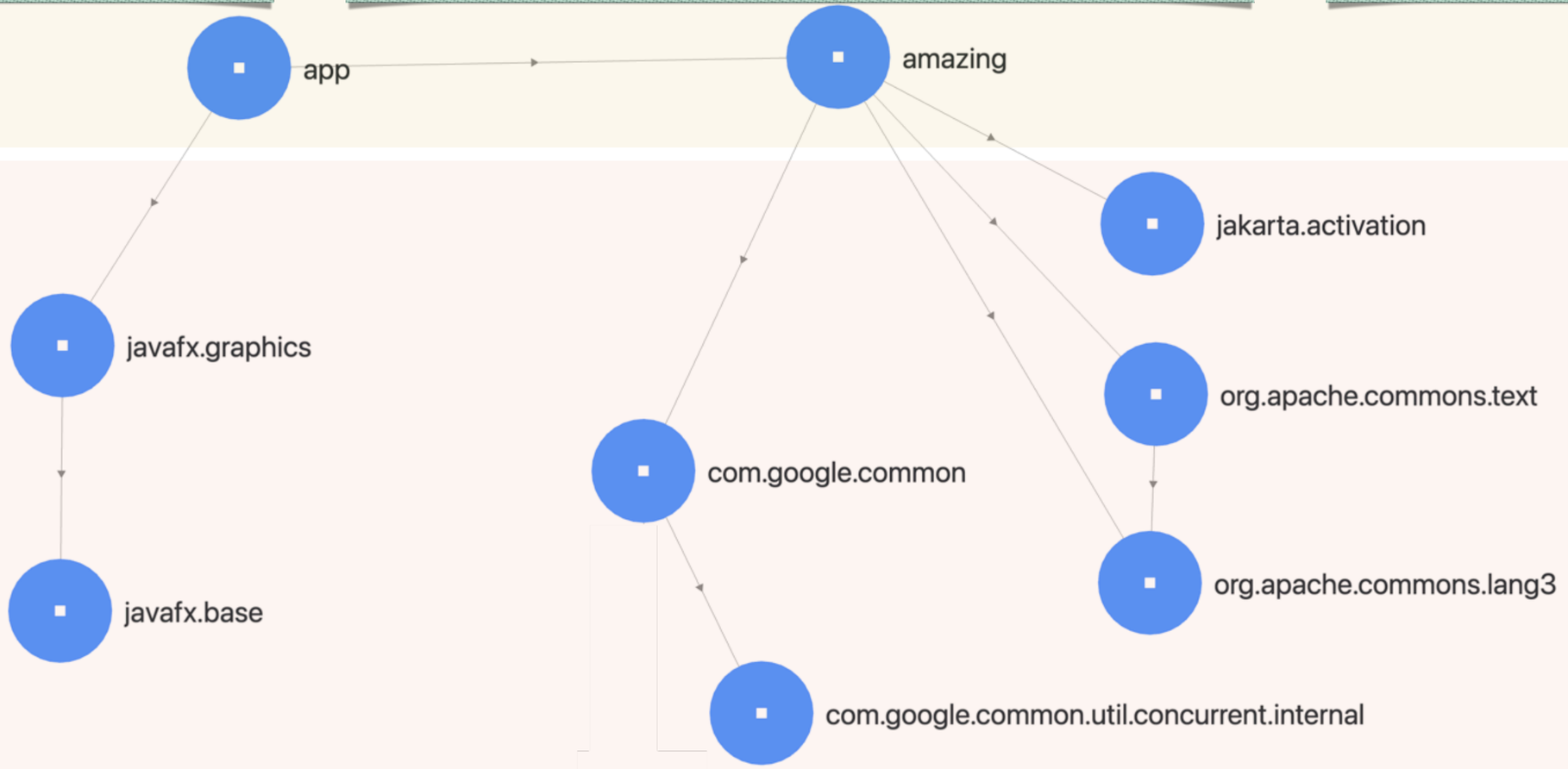
Scopes are important

Essential Complexity #2

Multiple **Versions/Variants** of 3rd party Modules

"Essential" Complexity #3

The world is **not perfect**



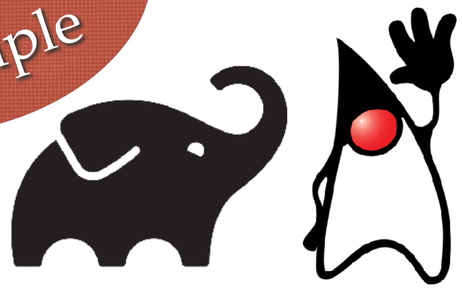
Dedicated places for dependency definitions

Unified syntax for scope definition

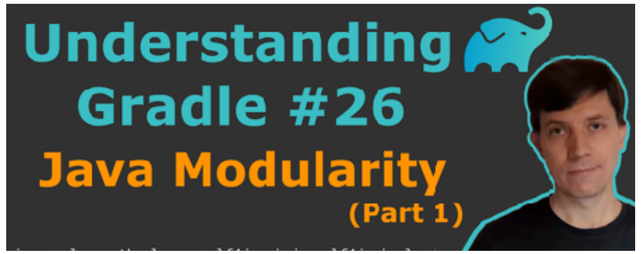
Variant-aware conflict detection and resolution (as provided in Gradle)

Central patch file(s) to enrich an fix metadata

slides & sample



https://github.com/jjohannes/java-module-system



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